UNIT 5

EFFICIENT DATA HANDLING AND PROTECTION IN MS EXCEL

Learning Objectives

- Apply **sorting and filtering** to manage large datasets.
- Understand the purpose and use of **conditional formatting** in MS Excel.
- Use different **worksheet functions** effectively.
- Perform **basic and advanced formatting** techniques.
- Protect **workbooks and worksheets** from unauthorized changes.
- Work with **multiple sheets**, including inserting, renaming, and deleting.
- Apply **data validation rules** in Excel.
- Implement **formula-based formatting** to highlight specific values.

Microsoft Excel is an essential tool for managing, analyzing, and organizing data effectively. Whether you're working with large datasets, formatting information for better clarity, or trying to ensure your spreadsheets remain secure, Excel offers a range of powerful features to streamline your workflow. In this guide, we will explore key functions including sorting, filtering, conditional formatting, and workbook protection, empowering you to enhance your data management skills and maintain organized, professional spreadsheets.

Sorting in MS Excel

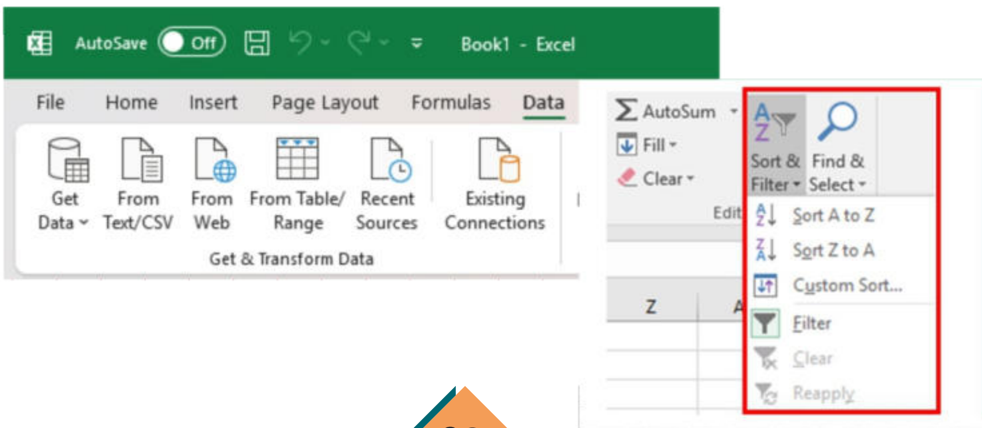
Sorting options can also be found on the Home tab, under the **Sort & Filter** command.

To sort alphabetically:

1. Select a cell in the column containing text data.
2. From the Data tab, click **Sort Ascending (A→Z)** to arrange the text from A to Z or **Sort Descending (Z→A)** to arrange it from Z to A.
3. The text entries will be sorted in alphabetical order.

To sort in numerical order:

1. Select a cell in the column you want to sort by (e.g., in our example, **C6**).
2. From the **Data** tab, click the **Sort Ascending (A→Z)** command to sort from smallest to largest, or the **Sort Descending (Z→A)** command to sort from largest to smallest.
3. The data in the spreadsheet will be organized numerically.



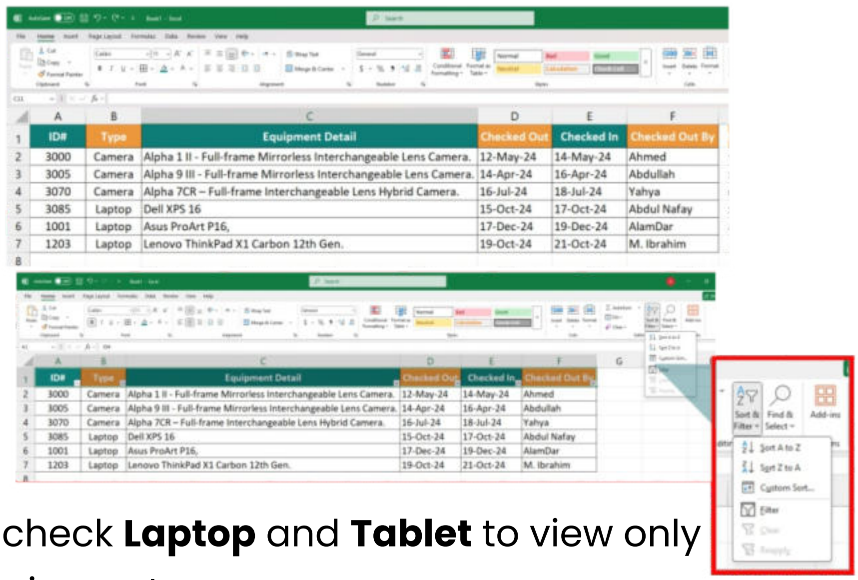
Filters in MS Excel

If your worksheet contains a lot of content, it can be difficult to find information quickly. **Filters** allow you to narrow down the data in your worksheet, making it easier to view only the information you need.

Filtering Data:

In the following example, a filter is applied to an equipment log worksheet to display only the laptops and projectors available for checkout.

1. Ensure that your worksheet includes a **header row** to identify column names. In this example, the worksheet is organized into different columns identified by header cells in row 1: **ID#, Type, Equipment Detail, etc.**
2. Select the **Data** tab, and click the **Filter** command.
3. A drop-down arrow ▼ will appear in each header cell.
4. Click the drop-down arrow in the column you want to filter. In this example, we filter column **B** to display only certain types of equipment.
5. The **Filter menu** will appear.
6. Uncheck the **Select All** box to quickly deselect all options.
7. Check the boxes next to the data you want to filter, then click **OK**. In this example, we check **Laptop** and **Tablet** to view only those types of equipment.



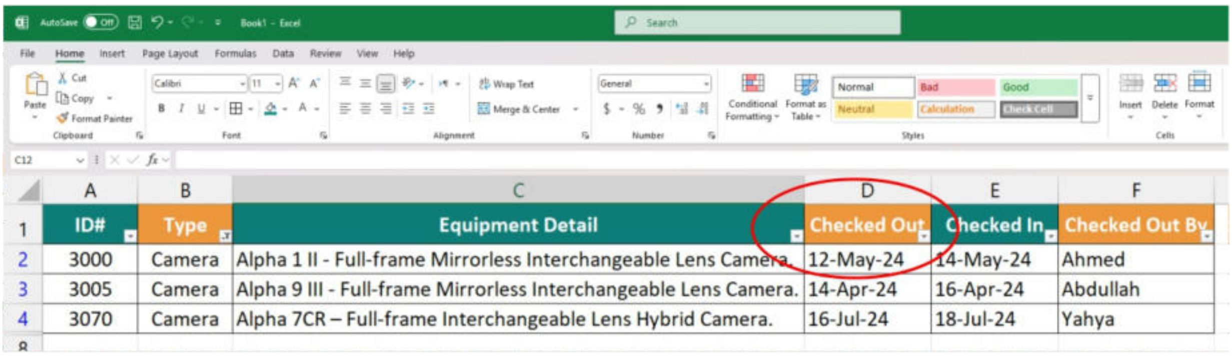
ID#	Type	Equipment Detail	Checked Out	Checked In	Checked Out By
3000	Camera	Alpha 1 II - Full-frame Mirrorless Interchangeable Lens Camera.	12-May-24	14-May-24	Ahmed
3005	Camera	Alpha 9 III - Full-frame Mirrorless Interchangeable Lens Camera.	14-Apr-24	16-Apr-24	Abdullah
3070	Camera	Alpha 7CR – Full-frame Interchangeable Lens Hybrid Camera.	16-Jul-24	18-Jul-24	Yahya
3085	Laptop	Dell XPS 16	15-Oct-24	17-Oct-24	Abdul Nafay
1001	Laptop	Asus ProArt P16,	17-Dec-24	19-Dec-24	AlamDar
1203	Laptop	Lenovo ThinkPad X1 Carbon 12th Gen.	19-Oct-24	21-Oct-24	M. Ibrahim

8. The data will be filtered, temporarily hiding any content that does not match the criteria. In this example, only **Laptops and Tablets** are visible.

Applying Multiple Filters

Filters are **cumulative**, meaning you can apply multiple filters to refine your results. In this example, the worksheet is already filtered to show **Laptops** and **Projectors**, and we want to further filter by date to show only items checked out in **August**.

1. Click the drop-down arrow in the column you want to filter. In this case, column **D (Checked Out Date)**.
2. The **Filter menu** will appear.
3. Check or uncheck the boxes to select the relevant dates. In this example, we uncheck all except **August** and click **OK**.
4. The new filter will be applied. Now, the worksheet displays only **Laptops and Tablets** checked out in **August**.



ID#	Type	Equipment Detail	Checked Out	Checked In	Checked Out By
3000	Camera	Alpha 1 II - Full-frame Mirrorless Interchangeable Lens Camera	12-May-24	14-May-24	Ahmed
3005	Camera	Alpha 9 III - Full-frame Mirrorless Interchangeable Lens Camera.	14-Apr-24	16-Apr-24	Abdullah
3070	Camera	Alpha 7CR – Full-frame Interchangeable Lens Hybrid Camera.	16-Jul-24	18-Jul-24	Yahya

How to Clear a Filter

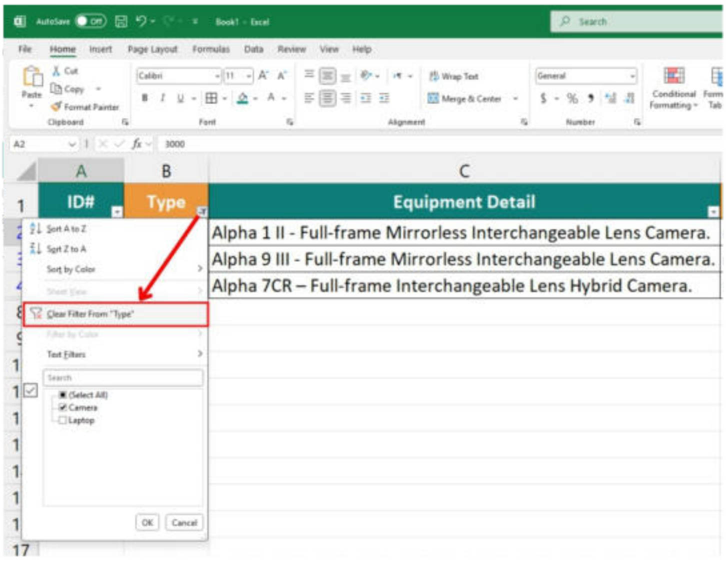
After applying a filter, you may want to remove or clear it to view the entire Dataset again.

1. Click the **drop-down arrow** in the column where the filter is applied. In this example, column **D (Checked Out Date)**.

2. The **Filter menu** will appear.

3. Select **Clear Filter from [COLUMN NAME]**. Here, we choose **Clear Filter from "Checked Out"**.

4. The filter will be removed, and the previously hidden data will be displayed again.



Conditional Formatting in MS Excel:

Excel **Conditional Formatting** is a powerful feature that helps highlight key information in your spreadsheets and identify variations in cell values at a glance. Conditional formatting in Excel is easy to use, and we will explore it with an example.

Finding Conditional Formatting in MS Excel:

For a start, let's locate the **Conditional Formatting** feature in the latest version of Microsoft Excel. It is found under the **Home tab → Styles group**.

Applying Conditional Formatting

1. In your Excel spreadsheet, select the cells you want to format.

○ In this example, we have a table listing **marks obtained by students out of 40**.

○ We aim to highlight the cells (**C5:C22**) containing marks **less than 13**.

2. Select the **cells C5:C22**.

3. Go to the **Home** tab, navigate to the **Styles** group, and click **Conditional Formatting**.

○ A menu will appear showing different formatting options like **Data Bars,Colour Scales, and Icon Sets**.

4. Since we need to apply conditional formatting for numbers **less than 13**, choose **Highlight Cells Rules → Less Than...**.

5. Enter the value in the dialog box under **Format cells that are LESS THAN**.

○ In our case, type **13** and choose a formatting style (e.g., red fill with dark red text).

6. Click **OK**. The selected range will now be highlighted based on the rule.

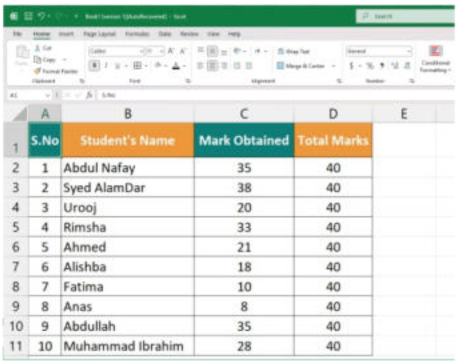
Other conditional formatting options include:

• Format values **greater than, less than, or equal to** a specific number.

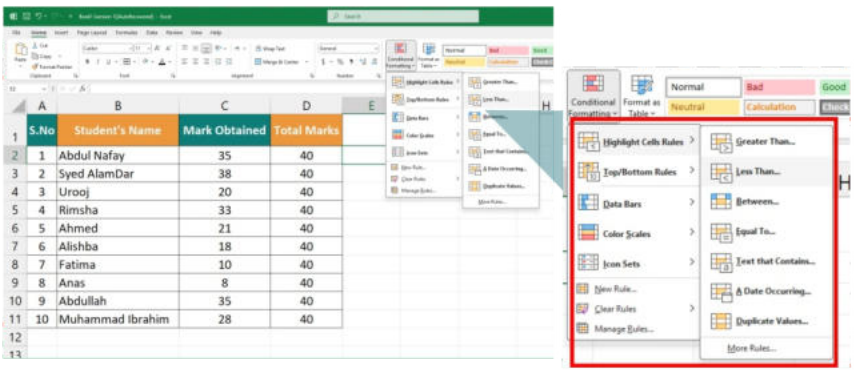
• Highlight text containing **specific words or characters**.

• Highlight **duplicate values**.

• Format **specific dates**.



S.No	Student's Name	Mark Obtained	Total Marks
1	Abdul Nafay	35	40
2	Syed AlamDar	38	40
3	Urooj	20	40
4	Rimsha	33	40
5	Ahmed	21	40
6	Alishba	18	40
7	Fatima	10	40
8	Anas	8	40
9	Abdullah	35	40
10	Muhammad Ibrahim	28	40



S.No	Student's Name	Mark Obtained	Total Marks
1	Abdul Nafay	35	40
2	Syed AlamDar	38	40
3	Urooj	20	40
4	Rimsha	33	40
5	Ahmed	21	40
6	Alishba	18	40
7	Fatima	10	40
8	Anas	8	40
9	Abdullah	35	40
10	Muhammad Ibrahim	28	40

Working with Worksheets in Excel

Understanding a Worksheet:

A **worksheet** is a collection of **cells** where data is stored and manipulated. Each Excel workbook contains **multiple worksheets** by default.

- A worksheet begins with **row number 1** and **column A**.
- The maximum size of a worksheet is **1,048,576 rows × 16,384 columns**.

Selecting a Worksheet:

When you open Excel, **Sheet1** is automatically selected.

- The name of the worksheet appears in a **tab at the bottom** of the window.
- To switch between sheets, click on **Sheet2 or Sheet3**.

Renaming a Worksheet:

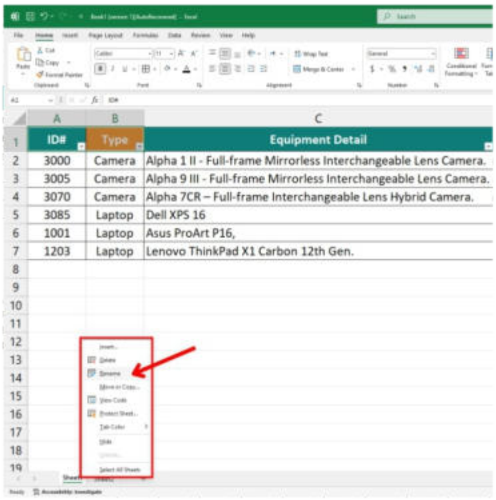
By default, Excel names worksheets **Sheet1, Sheet2, Sheet3**, etc.

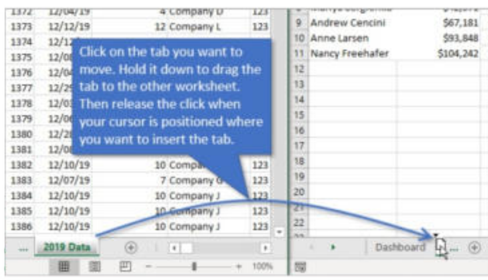
To rename a worksheet:

1. Right-click on the sheet tab (e.g., Sheet1).

2. Click **Rename**.

3. Type a new name, e.g., **Class VI**.





Managing Worksheets

Inserting a Worksheet:

You can insert as many worksheets as needed.

• To insert a new worksheet quickly, **click the Insert Worksheet tab** at the bottom of the document window.

Moving a Worksheet:

To rearrange worksheets:

1. Click and **drag** the sheet tab (e.g., Sheet4) to its new position.
2. For example, drag **Sheet4** before **Sheet 2**.

Deleting a Worksheet:

To delete a worksheet:

1. Right-click on a sheet tab.

2. Select **Delete** (e.g., delete **Sheet4**).

Copying a Worksheet

Instead of recreating a worksheet manually, you can duplicate it.

Steps to Copy a Worksheet:

1. **Right-click** on the sheet tab (e.g., **Class VI**).
2. Choose **Move or Copy....**
3. Select **(move to end)** and check **Create a copy**.
4. Click **OK**.
5. Rename the new worksheet (e.g., **Class VI Games**).

Protecting Workbooks and Worksheets

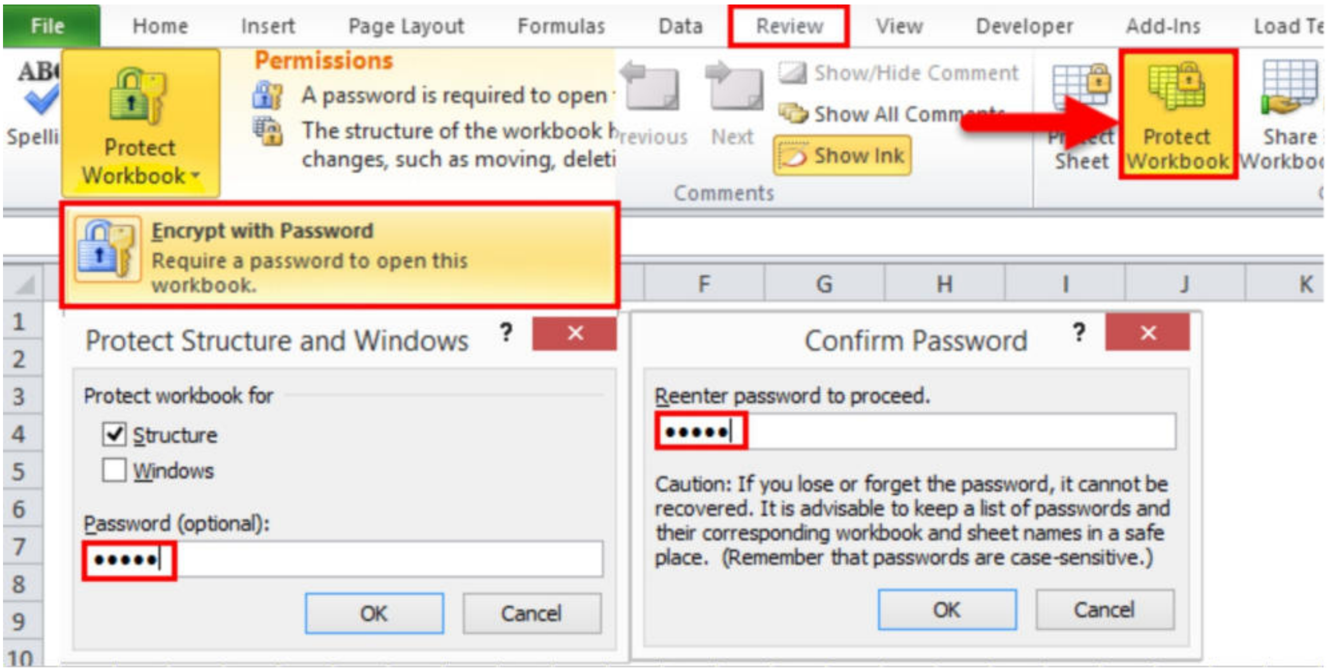
Protecting a Workbook:

You can protect a workbook to **prevent changes** to its structure.

1. Open the **Review** tab and click **Protect Workbook**.
2. Check **Structure**, enter a **password**, and click **OK**.

3. Re-enter the password and confirm.

4. Once applied, users **cannot insert, delete, rename, move, copy, hide, or unhide worksheets**.



Protecting Worksheet Windows:

Protecting worksheet windows prevents users from **moving, resizing, or closing** them.

1. Open the **Review** tab and click **Protect Workbook**.

2. Check **Windows**, enter a **password**, and click **OK**.

Protecting a Worksheet:

To prevent unauthorized changes:

1. Right-click on a worksheet tab.

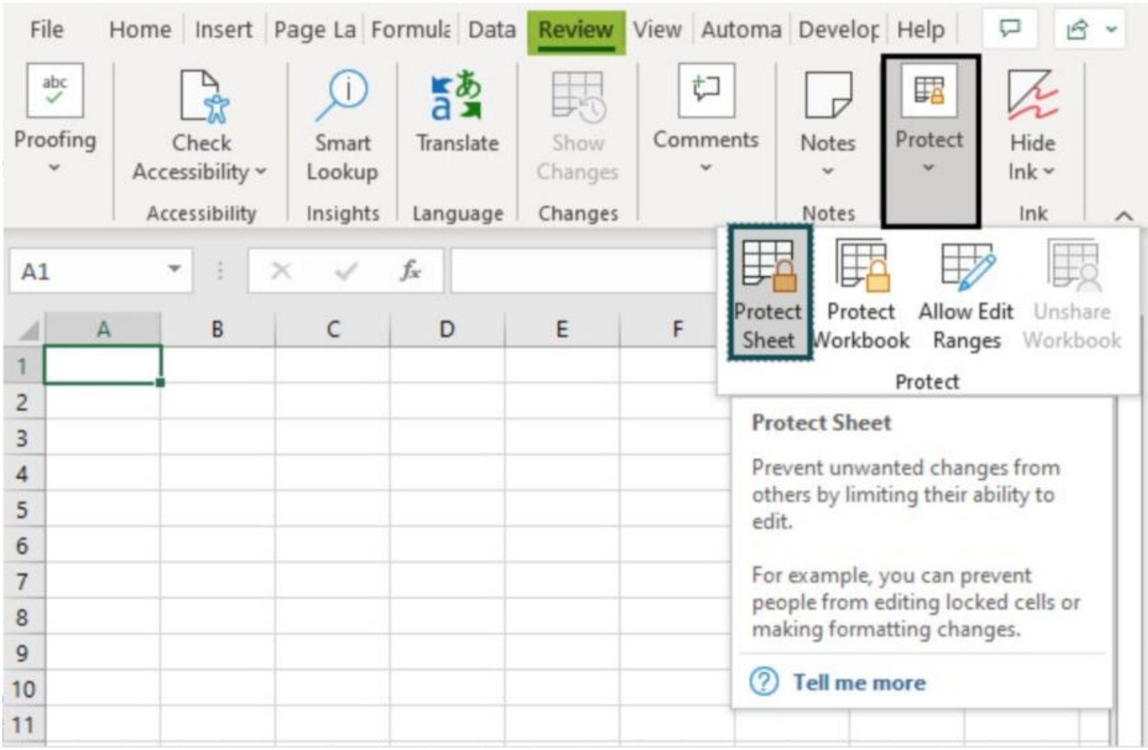
2. Select **Protect Sheet**.

3. Enter a **password**.

4. Select the actions users **are allowed** to perform.

5. Click **OK** and confirm the password.

To unprotect the worksheet, **right-click the tab and select Unprotect Sheet**.



Protect Sheet

EXERCISES

01: Choose the correct answer.

1. What is the primary function of conditional formatting?

a) To delete cells

b) To change cell colours based on conditions

c) To sort data

d) To create new worksheets

---

2. How do you access conditional formatting in Excel?

a) Insert tab

b) Data tab

c) Home tab → Styles group

d) View tab

---

3. Which function highlights duplicate values?

a) Data Validation

b) Conditional Formatting

c) IF Function

d) VLOOKUP

4.How can you protect a worksheet in Excel?

a) By clicking "Hide Sheet"

b) By using "Protect Sheet" under the Review tab

c) By deleting unwanted rows

d) By saving it in a different format

---

5. What does the NOW( ) function return?

a) Current Date and Time

b) Sheet Name

c) Cell Address

d) The Last Saved Time

---

6. The Sort & Filter feature is found under which tab?

a) Insert

b) Home

c) Data

d) View

7. How can you rename a worksheet?

a) Double-click on the sheet tab

b) Right-click and select "Rename"

c) Use the Format option under the Home tab

d) All of the above

8. What is the shortcut key to apply filters in Excel?

a) Ctrl + P

b) Ctrl + Shift + L

c) Ctrl + D

d) Ctrl + S

9. What is the maximum number of rows in an Excel worksheet?

a) 104,8576

b) 1,048,576

c) 65,536

d) Unlimited

10. What happens when you click "Clear Filter" on a column?

a) All data is deleted

b) Filtering is removed, and all data is displayed

c) Only selected rows are deleted

d) Data is sorted automatically

02: Fill in the blanks.

1. Conditional formatting helps to highlight ___________ values in a spreadsheet.

2. The shortcut key to apply filters is ___________.

3. In Excel, the default number of worksheets in a new workbook is ___________.

4. The ___________ tab contains the Sort & Filter feature.

5. The function **=NOW( )** returns ___________.

6. You can rename a worksheet by ___________ on its tab.

7. The **Protect Workbook** option is found under the ___________ tab.

8. The **AutoSum** button is used to quickly calculate ___________.

9. ___________ is used to highlight duplicate values in Excel.

10. Excel formulas always start with the ___________ symbol.

04: Answer the following questions briefly:

1. What is sorting in MS Excel?
2. How does the filter function help in Excel?
3. What is the use of multiple filters in Excel?
4. Which tab contains the "Filter" option in MS Excel?
5. What is conditional formatting?
6. What is the default colour used in Excel when applying conditional formatting?

05: Provide detailed answers of the following questions:

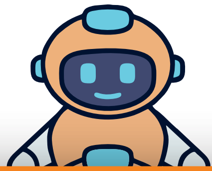
1. Explain the process of sorting data in Excel with an example.
2. What is filtering data in Excel? Describe how it is applied.
3. What is the difference between filtering and sorting in Excel?
4. Explain how to apply multiple filters in a worksheet.

06: Viva Questions:

1. What is the difference between absolute and relative cell references?
2. How do you remove a filter in Excel?
3. What happens when you hide a worksheet?

4. Can you apply multiple conditional formatting rules to the same cell?
5. How do you clear all formatting from a worksheet?
6. Explain the purpose of the IF function in Excel.
7. How can you highlight text-based values using conditional formatting?
8. What are the advantages of using data validation in Excel?
9. How can you lock specific cells while allowing editing in other parts of the worksheet?

Lab Activities:



1. Apply Conditional Formatting: Create a student marksheet and highlight all scores below 40 in red.

2. Sorting and Filtering: Given a dataset of 20 students, sort them in alphabetical order and apply a filter to show only those who scored above 70.

3. Worksheet Management: Create a workbook with three worksheets (Attendance, Marks, Reports). Rename them and format the header row differently for each sheet.

4. Protecting a Sheet: Create an employee salary sheet, protect it with a password, and allow only specific cells to be edited.